University of Hertfordshire
The School of Physics Engineering & Computer Science
Data Science Subject Group

Instructor: Dr. Yi Sun
Release Date: 25 March 2026

Image Classification Challenge

Call for Competition

Competition Objective

This year's Image Classification Club competition focuses on building a prototype system for Facial Expression Classification using deep learning models. Participants will explore how computer vision models can be used to recognise and classify facial expressions from images.

Dataset

The dataset for this competition is FER-2013 (Facial Expression Recognition 2013) and can be accessed at the following link:

https://www.kaggle.com/datasets/msambare/fer2013/data

The dataset contains labelled grayscale images of faces representing different emotional expressions. The seven expression classes are: Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral.

Tasks

1. Image Classification
   1) Explore the dataset and understand the structure, class distribution, and characteristics of the images.
   2) Use the provided training and test sets. Clearly explain the data preprocessing steps.
   3) Implement at least three image classification models. Two models from TensorFlow Keras must be common across all groups for fair comparison:
      - ResNet50V2 (resnet_v2_50_imagenet) transfer learning model
      - Vision Transformer (ViT) transfer learning model

      The third model can be chosen by participants (e.g., EfficientNet, MobileNet, etc.).
   4) Monitor training performance using learning curves to analyse model behaviour on both training and validation sets.
   5) Evaluate all models using the same test set.
   6) Report the following metrics: accuracy, confusion matrix, and related performance metrics (recall, precision, F1-score).
   7) Analyse and discuss model errors and limitations.

2. Responsible Deployment: System Design, Temporal Smoothing, Data Protection, and Ethics

The FER-2013 dataset contains individual images, while real-world facial expression recognition systems often operate on video streams and must be designed as complete systems rather than as standalone models. In practice, such systems may involve a frontend, middleware, and backend architecture, and predictions may fluctuate between frames due to noise, lighting changes, or small variations in input. This can lead to unstable or inconsistent predictions over time.

For this competition, participants are not required to implement a full deployment system. Instead, include a short discussion in the submitted poster explaining how the model could be used in a responsible real-world system. Each poster should also include a small system diagram showing the frontend, middleware, and backend components. The diagram should illustrate how these components interact and suggest which software tools or toolkits could be used to implement each component.

The discussion should address:

- What the frontend, middleware, and backend of such a system might do.
- How temporal smoothing can reduce frame-to-frame fluctuations and produce more stable, reliable predictions over time.
- How facial images relate to privacy, personal data, and ethical concerns, including whether they may be considered biometric data under general data protection regulation (GDPR) and what this implies for real-world deployment.

Participants should explain ideas clearly and discuss what practical challenges might arise in deploying such a system.

Before writing this section, participants should read the UH Ethics Approval guidance and data protection guidance on Canvas:
https://herts.instructure.com/courses/105037
https://herts.instructure.com/courses/105037/pages/data-protection-what-do-i-need-to-consider

Optional Personal Testing

Participants may optionally test their models using their own images or short video recordings.
Testing may be performed on a single image, a sequence of images, or a short video. Any of these options are acceptable.

The following rules must be followed:

- Only self-data may be used. Each participating student may use images or videos of themselves only.
- All personal data must remain local and must not be submitted. Participants should also follow basic data protection practices, including secure storage (e.g. password-protected devices), limiting access, and deleting data after use, to reduce risks such as unauthorised access or loss of devices.

- Posters must only include aggregated results (e.g., accuracy, confusion matrices).
- No personal images or videos should appear in the poster or submission.

Teamwork

Students are encouraged to participate in teams of up to four members. Individual participation is also allowed.

Poster Presentation

Each team or individual participant will present their work at a conference-style poster session.

Evaluation Criteria

- Model performance on the test dataset
- Technical approach and innovation
- Critical reflection and discussion of limitations
- Quality of the poster and clarity of explanation

Submission

Each team must submit the following:

- A Jupyter Notebook containing code and experiments
- A poster (PDF format) summarising the project

Important Dates

- Competition registration opens: To be announced (week commencing 15/04/2026)
- Notebook and poster submission deadline: To be announced (week commencing 18/05/2026)
- Final poster demonstration session: To be announced (in-person on campus) (week commencing 25/05/2026 or week commencing 01/06/2026)

We look forward to your creative ideas and innovative solutions!